

ITSimplera Institute



Week 1 Task Report

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Network Administration

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Introduction

This lab focused on IP subnetting using Fixed Length Subnet Masking (FLSM) and Variable Length Subnet Masking (VLSM), as well as Virtual Local Area Network (VLAN) implementation. The objective was to design logical network segments, assign IP addresses efficiently, configure VLANs, establish trunk links between switches, and verify network connectivity.

Fixed Length Subnet Masking (FLSM)

FLSM divides a network into multiple subnets of equal size. Each subnet uses the same subnet mask, providing a simple and structured addressing scheme. FLSM is suitable when all network segments require a similar number of hosts.

Variable Length Subnet Masking (VLSM)

VLSM allows different subnet masks within the same network. This enables efficient use of IP addresses by allocating larger subnets to large departments and smaller subnets to smaller departments. VLSM reduces address wastage and improves scalability.

VLAN (Virtual Local Area Network)

A VLAN is a logical segmentation of a switched network. Devices in the same VLAN can communicate as if they are on the same physical network, regardless of location. VLANs improve security, reduce broadcast traffic, and simplify network management.

Trunk Links

A trunk link carries traffic for multiple VLANs between switches. IEEE 802.1Q tagging is used to identify VLAN traffic across trunk ports. In this lab, VLAN 10 (HR) and VLAN 20 (IT) were allowed across the trunk connection between switches.

Objectives

- Understand FLSM and VLSM subnetting concepts.
- Configure VLANs on Cisco switches.
- Establish trunk communication between switches.
- Verify VLAN operation using CLI commands and ping tests.

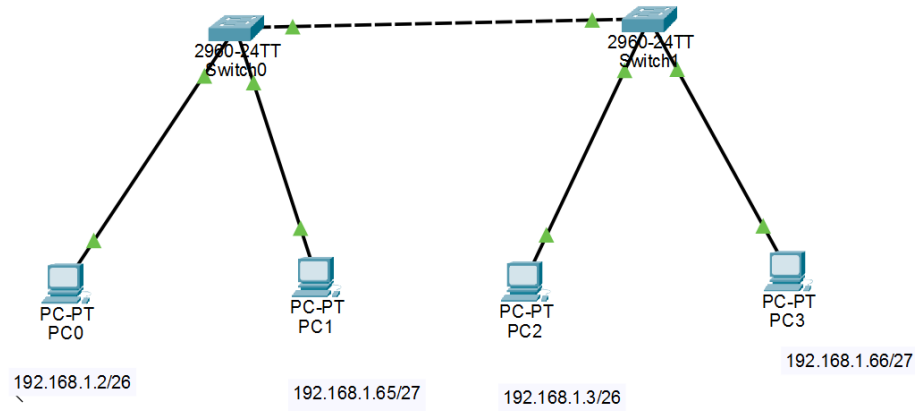
Tools Used

- GNS3
- PuTTY
- Cisco Packet Tracer

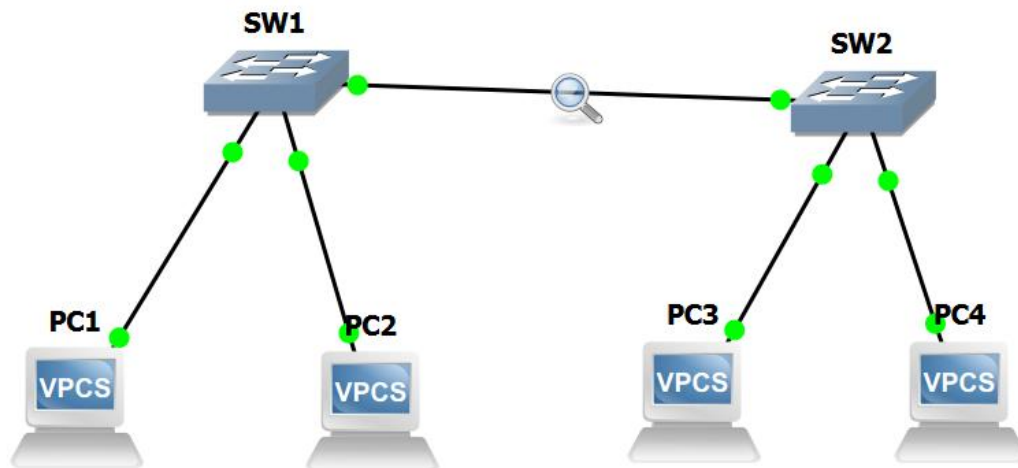
Network Design and Screenshots

Cisco Packet Tracer Topology

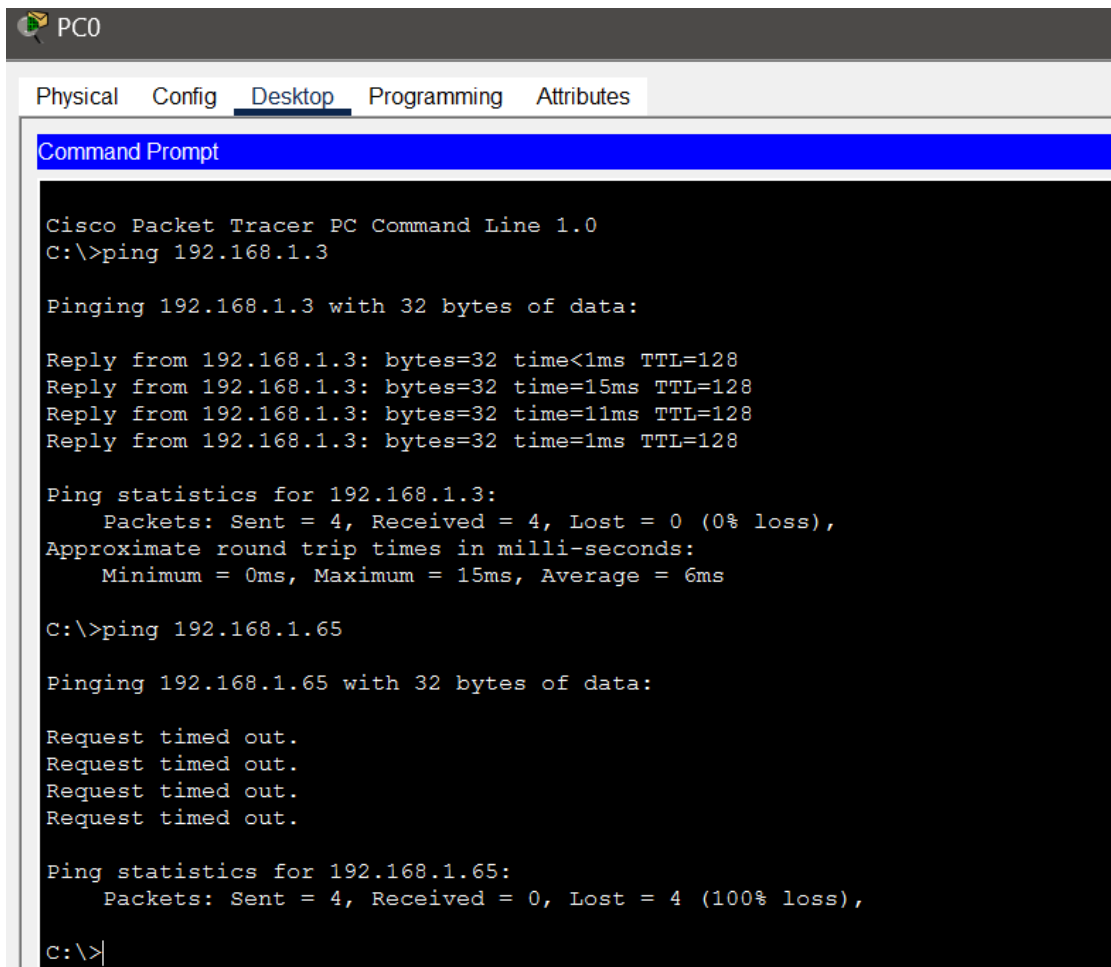
Setting up this exact FLSM/VLSM and VLAN lab in Cisco Packet Tracer



GNS3 Topology



PC0 Ping Verification



The image shows a screenshot of the Cisco Packet Tracer interface for PC0. The 'Desktop' tab is selected, and the 'Command Prompt' window is open. The command prompt displays the results of two ping commands. The first command, 'ping 192.168.1.3', shows successful results with four replies and 0% loss. The second command, 'ping 192.168.1.65', shows four request timeouts and 100% loss.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time=15ms TTL=128
Reply from 192.168.1.3: bytes=32 time=11ms TTL=128
Reply from 192.168.1.3: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 15ms, Average = 6ms

C:\>ping 192.168.1.65

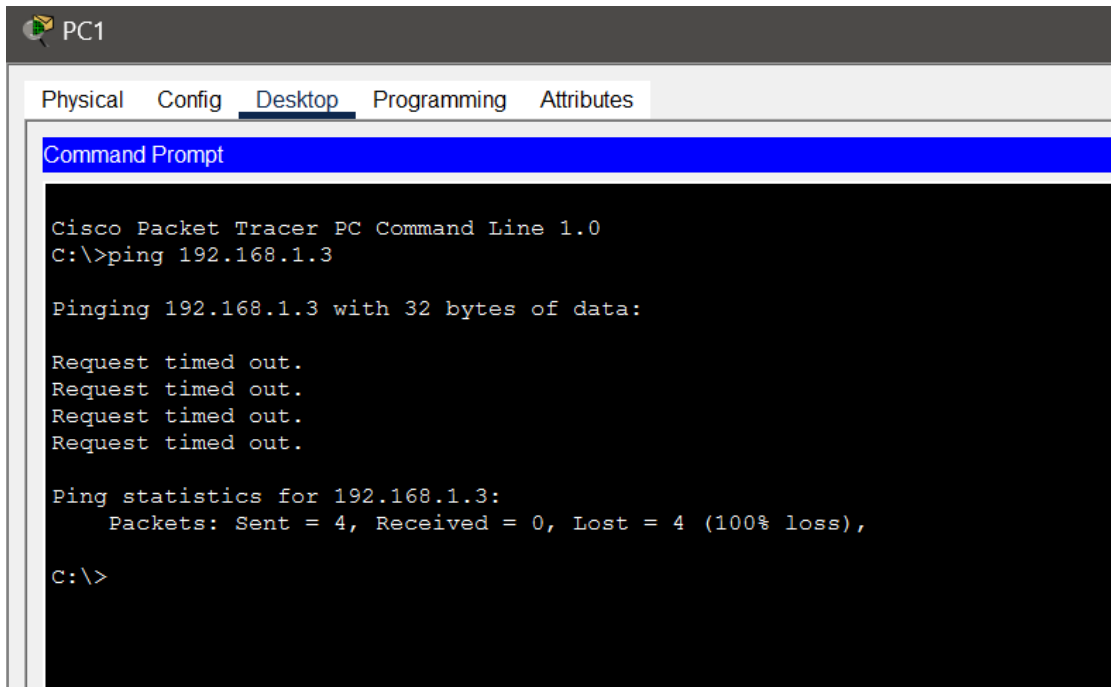
Pinging 192.168.1.65 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.1.65:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>|
```

PC1 Ping Verification



The screenshot shows the PC1 interface in Cisco Packet Tracer. The 'Desktop' tab is selected, and the 'Command Prompt' application is open. The command prompt displays the output of a ping command to 192.168.1.3, showing four failed requests and a 100% loss rate.

```
PC1
Physical Config Desktop Programming Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.3

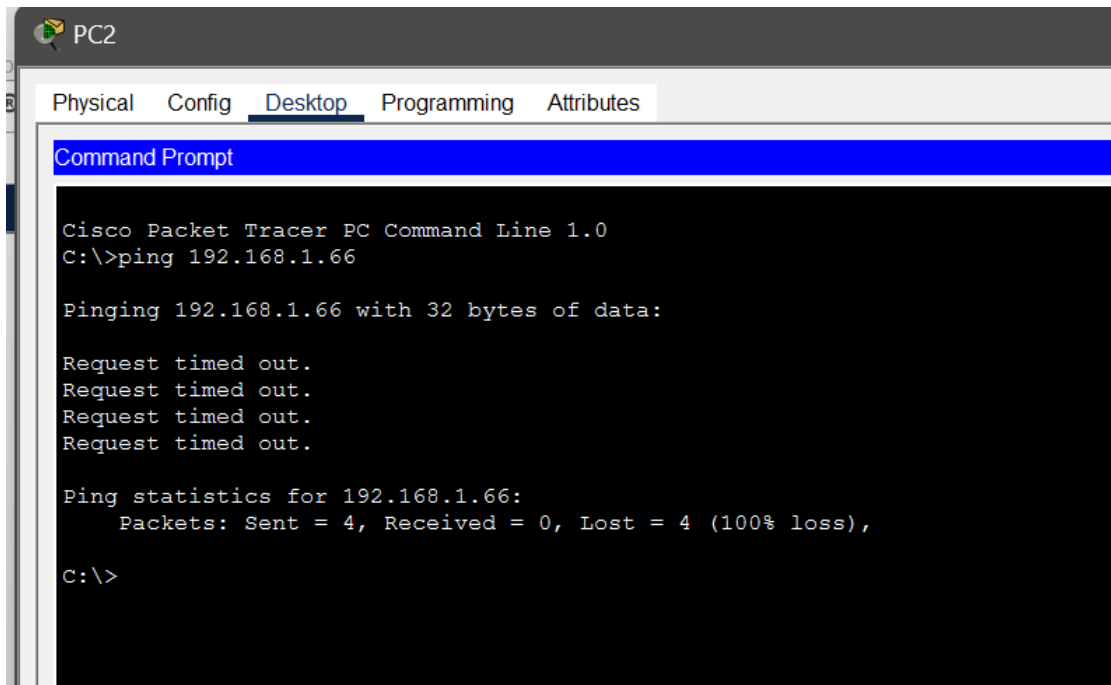
Pinging 192.168.1.3 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

PC2 Ping Verification



The screenshot shows the PC2 interface in Cisco Packet Tracer. The 'Desktop' tab is selected, and the 'Command Prompt' application is open. The command prompt displays the output of a ping command to 192.168.1.66, showing four failed requests and a 100% loss rate.

```
PC2
Physical Config Desktop Programming Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.66

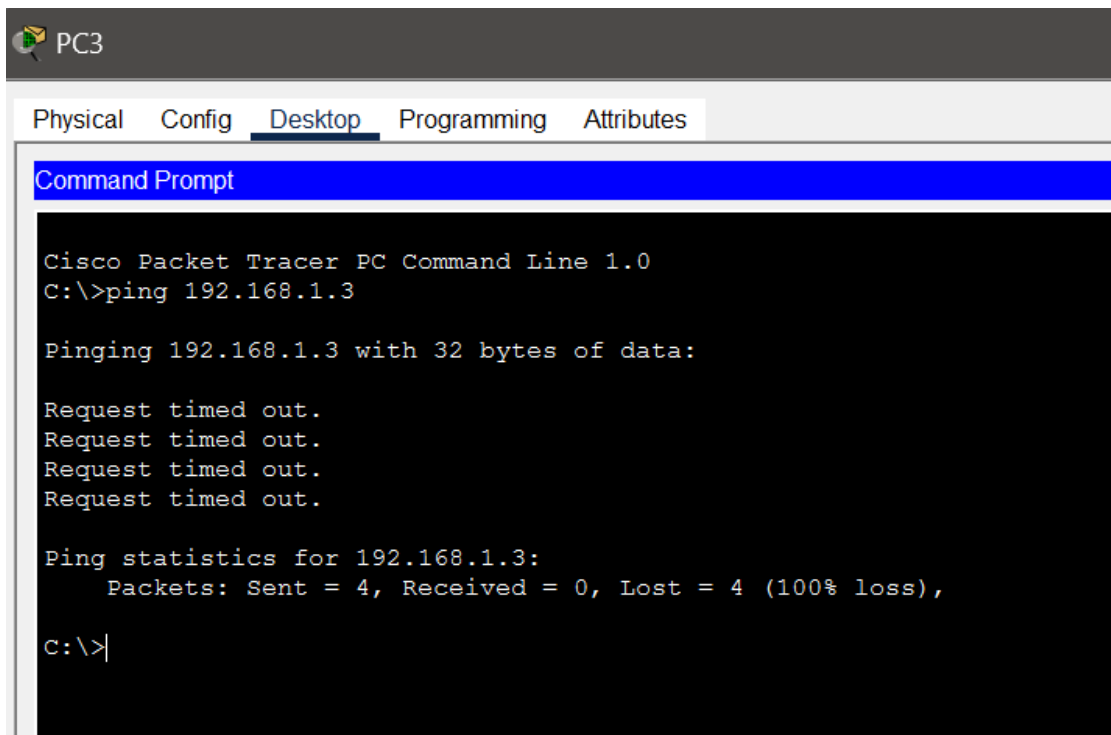
Pinging 192.168.1.66 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.1.66:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

PC3 Ping Verification



The screenshot shows the PC3 interface with the Desktop tab selected. A Command Prompt window is open, displaying the output of a ping command to 192.168.1.3. The output shows four 'Request timed out' messages and a summary indicating 100% loss of packets.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>|
```

SW1 VLAN Brief Output

```
SW1#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/3, Fa0/4, Fa0/5, Fa0/6 Fa0/7, Fa0/8, Fa0/9, Fa0/10 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/1, Gig0/2
10	HR	active	Fa0/2
20	IT	active	Fa0/11
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

SW2 VLAN Brief Output

```
SW2#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/3, Fa0/4, Fa0/5, Fa0/6 Fa0/7, Fa0/8, Fa0/9, Fa0/10 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/1, Gig0/2
10	HR	active	Fa0/2
20	IT	active	Fa0/11
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
SW2#
```

SW1 Trunk Verification

```
SW1#  
SW1#show interface trunk
```

Port	Mode	Encapsulation	Status	Native vlan
Fa0/1	on	802.1q	trunking	1


```
Port Vlan allowed on trunk  
Fa0/1 10,20
```



```
Port Vlan allowed and active in management domain  
Fa0/1 10,20
```



```
Port Vlan in spanning tree forwarding state and not pruned  
Fa0/1 10,20
```

SW2 Trunk Verification

```
SW2#  
SW2#show interface trunk
```

Port	Mode	Encapsulation	Status	Native vlan
Fa0/1	on	802.1q	trunking	1


```
Port Vlan allowed on trunk  
Fa0/1 10,20
```



```
Port Vlan allowed and active in management domain  
Fa0/1 10,20
```



```
Port Vlan in spanning tree forwarding state and not pruned  
Fa0/1 10,20
```

Configuration Summary

VLAN 10 (HR) and VLAN 20 (IT) were created on both switches. Access ports were assigned to their respective VLANs. A trunk link was configured between SW1 and SW2 using 802.1Q encapsulation. Hosts were assigned IP addresses based on the subnetting design.

Verification and Results

The commands "show vlan brief" and "show interface trunk" confirmed successful VLAN creation and trunk operation. Ping tests demonstrated successful communication between devices belonging to the same VLAN and expected isolation between devices in different VLANs.

Challenges Faced

During configuration, attention was required to correctly assign access ports, configure trunk links, and verify VLAN membership. Connectivity issues were resolved through VLAN and trunk verification.

Learning Outcomes

This lab enhanced understanding of subnetting techniques, VLAN deployment, switch configuration, trunking concepts, and network troubleshooting procedures.

Conclusion

The objectives of the Week 2 internship task were successfully achieved. The lab demonstrated the practical implementation of FLSM, VLSM, VLANs, trunking, and connectivity verification in both Cisco Packet Tracer and GNS3 environments.